

BUCURESTI BANEASA, Romania

WMO: 154200

Lat: 44.5106N

Lon: 26.0781E

Elev: 91

StdP: 100.23

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-10.1	-15.9	0.9	-11.9	-12.9	1.2	-8.8	10.3	-1.0	8.7	-0.8	1.9	250	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	34.4	21.0	32.8	20.6	31.2	20.0	22.7	30.3	21.9	29.5	21.1	28.4	2.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.6	24.7	19.8	14.7	23.9	19.0	13.9	23.3	67.5	30.4	64.4	29.5	61.7	28.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.5	5.5	DB	-18.5	36.8	3.4	2.0	-20.9	38.2	-23.0	39.4	-24.9	40.5	-27.4	41.9	(4)
(5)			WB	-18.7	24.5	3.4	1.4	-21.1	25.5	-23.1	26.3	-25.0	27.0	-27.5	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-1.3	1.3	6.1	11.7	16.9	21.0	23.0	22.8	17.7	11.5	6.1	0.4	(6)	
	DBStd	9.33	4.70	5.13	4.45	3.59	3.21	3.00	2.58	2.68	3.44	3.96	4.16	4.46	(7)	
	HDD10.0	1208	351	244	133	24	1	0	0	0	0	30	129	297	(8)	
	HDD18.3	2932	609	477	379	201	65	11	1	2	52	213	366	556	(9)	
	CDD10.0	1751	0	1	13	73	215	329	403	396	231	77	13	0	(10)	
	CDD18.3	432	0	0	0	1	21	89	146	140	33	2	0	0	(11)	
	CDH23.3	5253	0	0	1	44	368	1054	1683	1635	425	41	1	0	(12)	
	CDH26.7	2151	0	0	0	7	88	406	762	755	127	6	0	0	(13)	
(14) Wind	WSAvg	2.2	2.3	2.5	2.7	2.6	2.1	1.9	1.9	1.8	1.9	1.9	2.1	2.3	(14)	
(15) Precipitation	PrecAvg	631	41	35	40	49	69	77	64	54	54	44	47	47	(15)	
	PrecMax	1077	126	99	113	160	233	207	222	146	270	132	215	171	(16)	
	PrecMin	374	3	2	3	1	8	10	5	1	4	0	1	0	(17)	
	PrecStd	141	27	23	28	29	46	46	43	37	47	35	34	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	17.8	22.1	27.7	31.2	34.6	37.2	36.9	32.7	27.1	21.1	13.6	(19)	
		MCWB	7.7	11.0	12.5	16.4	18.7	21.2	21.5	21.3	19.8	17.6	14.1	9.1	(20)	
	2%	DB	8.9	14.2	19.5	24.2	29.0	32.7	34.6	34.5	29.9	24.0	17.9	10.3	(21)	
		MCWB	5.9	8.7	11.2	14.9	17.8	20.9	21.2	20.9	18.7	16.0	12.4	7.7	(22)	
	5%	DB	6.4	11.1	17.1	21.6	27.2	30.9	32.7	32.8	27.9	21.2	14.9	8.0	(23)	
		MCWB	5.0	7.1	10.0	13.5	17.2	20.2	20.8	20.5	17.8	14.7	11.1	6.2	(24)	
	10%	DB	4.6	8.3	14.6	19.5	25.2	29.0	30.9	31.0	25.8	18.9	12.4	6.1	(25)	
		MCWB	3.4	5.8	8.6	12.4	16.4	19.4	20.2	19.9	16.9	13.5	10.1	4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	11.5	13.3	17.2	20.8	23.6	23.9	24.2	21.4	18.9	14.8	10.2	(27)	
		MCDB	11.6	16.6	20.2	25.6	28.3	31.0	31.1	31.6	29.1	25.5	19.4	12.3	(28)	
	2%	WB	6.5	9.2	11.7	15.7	19.2	22.2	22.7	22.6	19.7	16.9	13.1	8.1	(29)	
		MCDB	8.1	13.4	18.4	22.5	26.2	29.7	30.4	30.0	26.9	21.8	16.5	10.1	(30)	
	5%	WB	5.1	7.6	10.4	14.4	18.1	21.2	21.9	21.6	18.8	15.6	11.8	6.5	(31)	
		MCDB	6.3	10.4	16.2	19.9	24.7	28.4	29.4	29.2	25.6	19.9	14.4	7.7	(32)	
	10%	WB	3.4	6.1	9.1	13.2	17.1	20.4	21.2	20.8	17.9	14.2	10.5	5.0	(33)	
		MCDB	4.4	8.2	13.4	18.0	23.5	27.1	28.3	28.2	23.8	17.8	12.2	6.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	9.1	11.2	12.5	13.7	13.5	13.9	14.3	13.2	11.6	8.7	7.1	(35)	
		MCDBR	10.2	14.4	16.6	17.0	17.4	16.6	17.1	17.6	17.0	16.0	13.9	10.5	(36)	
	5% WB	MCWBR	7.3	9.3	9.2	8.4	7.4	6.2	5.9	5.8	6.8	8.4	8.6	7.6	(37)	
		MCWBR	9.6	13.0	15.6	14.5	14.9	14.7	14.5	14.9	14.8	14.4	11.9	9.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.378	0.413	0.451	0.427	0.436	0.446	0.461	0.424	0.401	0.374	0.350	(40)	
		taud	2.277	2.194	2.155	2.075	2.223	2.231	2.211	2.164	2.233	2.268	2.293	2.291	(41)	
	Ebn at Noon	Ebn at Noon	747	787	811	811	846	838	823	794	792	754	707	699	(42)	
		Edh at Noon	85	111	132	155	138	137	138	140	121	102	84	77	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.37	2.21	3.43	4.70	5.87	6.34	6.42	5.73	4.15	2.70	1.48	1.13	(44)	
	RadStd	RadStd	0.21	0.34	0.37	0.48	0.43	0.44	0.42	0.32	0.31	0.33	0.19	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-123	+69	+39

Nomenclature: See separate page